

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-18397407})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}1884 + \mathbf{Z}(1639 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 4599352 = 0$, of discriminant -18397407 . Class group invariants $(40 \ 10 \ 5)$, class number 2000; the three stored generator ideals (HNF) are $\begin{pmatrix} 62 & 7 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1039 & 479 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)y^2 + (5s + 339709)y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 310y^8 - 98729y^7 + 899066y^6 - 48894653y^5 + 269 \\ & 0719747y^4 - 38077663276y^3 + 1275994253977y^2 - 11483843568669y \\ & + 676054462444509 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right)s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1364932699083 = 3^3 \cdot 6101 \cdot 8286029$:

$$\begin{pmatrix} 151659188787 & 92029726837 & 9034141479 & 16054093719 & 81654813653 & 53887090936 & 74422383738 & 55978661180 & 27157940536 & 53924866653 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 380 factors with signed exponents of absolute value up to 12563867541167585379; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)y^2 + (5s + 339709)y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 310y^8 - 98729y^7 + 899066y^6 - 48894653y^5 + 269 \\ & 0719747y^4 - 38077663276y^3 + 1275994253977y^2 - 11483843568669y \\ & + 676054462444509 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right)s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4136948743179 = 3 \cdot 7^3 \cdot 4020358351$:

$$\begin{pmatrix} 84427525371 & 32662630560 & 82642821275 & 50040169739 & 12392051524 & 68065394625 & 74336205091 & 8005853897 & 43250118945 & 72115595837 \\ 0 & 7 & 0 & 1 & 5 & 2 & 1 & 0 & 1 & 3 \\ 0 & 0 & 7 & 6 & 1 & 1 & 6 & 3 & 3 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)y^2 + (5s + 339709)y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 310y^8 - 98729y^7 + 899066y^6 - 48894653y^5 + 2690719747y^4 - 38077663276y^3 + 1275994253977y^2 - 11483843568669y + 676054462444509$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ 1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right)s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1364932699083 = 3^3 \cdot 6101 \cdot 8286029$:

$$\begin{pmatrix} 151659188787 & 92029726837 & 9034141479 & 16054093719 & 81654813653 & 53887090936 & 74422383738 & 55978661180 & 27157940536 & 53924866653 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 385 factors with signed exponents of absolute value up to 5407207007136419622; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 3 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 381y^3 + (-9s - 52315)y^2 + (-357s - 587577)y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 762y^8 - 104639y^7 - 1030350y^6 + 18803799y^5 + 3557747263y^4 + 99082836570y^3 + 2268632753133y^2 + 21700095645411y + 147963744385227$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right) s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1445520196512 = 2^5 \cdot 3^6 \cdot 7 \cdot 8852147$:

$$\begin{pmatrix} 3346111566 & 2214016038 & 1581849742 & 2259694972 & 3195991622 & 1600146428 & 536938132 & 2945984644 & 3191610299 & 2703188219 \\ 0 & 54 & 12 & 42 & 50 & 23 & 45 & 31 & 34 & 39 \\ 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 383 factors with signed exponents of absolute value up to 529906430669; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 381y^3 + (-9s - 52315)y^2 + (-357s - 587577)y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 762y^8 - 104639y^7 - 1030350y^6 + 18803799y^5 + 355774726 \\ & 3y^4 + 99082836570y^3 + 2268632753133y^2 + 21700095645411y + 147 \\ & 963744385227 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right) s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $205522872989 = 7 \cdot 67 \cdot 438215081$:

$$\begin{pmatrix} 205522872989 & 99708957569 & 82125197537 & 178481675936 & 76460983232 & 22964376023 & 153333158652 & 198282447692 & 119271531358 & 65204618837 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 384 factors with signed exponents of absolute value up to 586913612429; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 381y^3 + (-9s - 52315)y^2 + (-357s - 587577)y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 762y^8 - 104639y^7 - 1030350y^6 + 18803799y^5 + 355774726 \\ & 3y^4 + 99082836570y^3 + 2268632753133y^2 + 21700095645411y + 147 \\ & 963744385227 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right)s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2658403028583 = 3^3 \cdot 13597 \cdot 7241257$:

$$\begin{pmatrix} 886134342861 & 519340949357 & 778399074402 & 728222445557 & 449587310514 & 332301010764 & 549781219118 & 857932609108 & 622628702979 & 244313521912 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 377 factors with signed exponents of absolute value up to 200806330069; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 482)y^3 + (20s - 44079)y^2 + (35s + 713966)y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 949y^8 - 84278y^7 + 6612677y^6 - 151021673y^5 + 2 \\ & 786318041y^4 - 41877012822y^3 + 430627163226y^2 - 3203861858505y \\ & + 12564715259943 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right)s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2081607454119 = 3 \cdot 693869151373$:

$$\begin{pmatrix} 2081607454119 & 1835215672271 & 996348053807 & 1346122103099 & 1120149904008 & 1632683719257 & 1244827671335 & 1005661010630 & 314913697573 & 1306500860355 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 288 factors with signed exponents of absolute value up to 1725940362787; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 482)y^3 + (20s - 44079)y^2 + (35s + 713966)y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 949y^8 - 84278y^7 + 6612677y^6 - 151021673y^5 + 2 \\ & 786318041y^4 - 41877012822y^3 + 430627163226y^2 - 3203861858505y \\ & + 12564715259943 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right)s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3850075537 = 7 \cdot 11 \cdot 50000981$:

$$\begin{pmatrix} 3850075537 & 1434080588 & 1597579715 & 3833635359 & 257950722 & 3475155485 & 2580354442 & 2407147704 & 1276244565 & 3848950618 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 288 factors with signed exponents of absolute value up to 1106990938546; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s - 482)y^3 + (20s - 44079)y^2 + (35s + 713966)y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 - 949y^8 - 84278y^7 + 6612677y^6 - 151021673y^5 + 2 \\ & 786318041y^4 - 41877012822y^3 + 430627163226y^2 - 3203861858505y \\ & + 12564715259943 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right)s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9560434789675 = 5^2 \cdot 7^3 \cdot 11 \cdot 61^2 \cdot 27239$:

$$\begin{pmatrix} 639707915 & 364902244 & 223980176 & 540741395 & 95491380 & 576664442 & 362818452 & 4839848 & 212624880 & 161079383 \\ 0 & 427 & 203 & 58 & 140 & 283 & 69 & 145 & 336 & 27 \\ 0 & 0 & 35 & 10 & 1 & 24 & 24 & 25 & 10 & 22 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 291 factors with signed exponents of absolute value up to 1216145498765; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 641y^3 + (8s - 4)y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1282y^8 + 468729y^6 + 257277944y^4 + 33804751120y^2 + 923108293632$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right)s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3379297465600 = 2^8 \cdot 5^2 \cdot 7^3 \cdot 1539403$:

$$\begin{pmatrix} 3017229880 & 2265225060 & 722616516 & 2370831304 & 1481317588 & 629026663 & 1843216854 & 758218013 & 534913726 & 333008840 \\ 0 & 70 & 54 & 12 & 55 & 18 & 67 & 58 & 59 & 42 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 164 factors with signed exponents of absolute value up to 5923241; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 641y^3 + (8s - 4)y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1282y^8 + 468729y^6 + 257277944y^4 + 33804751120y^2 + 923108293632$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right)s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $63249306624 = 2^{10} \cdot 3^3 \cdot 7^2 \cdot 46687$:

$$\begin{pmatrix} 15686832 & 14612640 & 10998624 & 12818577 & 354075 & 9233536 & 2573589 & 6514586 & 9436875 & 13507881 \\ 0 & 84 & 14 & 81 & 81 & 55 & 28 & 24 & 71 & 64 \\ 0 & 0 & 4 & 3 & 1 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 & 1 & 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 2524112; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 641y^3 + (8s - 4)y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 1282y^8 + 468729y^6 + 257277944y^4 + 33804751120y^2 + 923108293632$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right)s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $22226745936 = 2^4 \cdot 3 \cdot 47^2 \cdot 209623$:

$$\begin{pmatrix} 118227372 & 110213778 & 103294564 & 65726847 & 101011218 & 48267241 & 108578627 & 63915267 & 24632636 & 3108525 \\ 0 & 94 & 0 & 93 & 84 & 0 & 74 & 67 & 39 & 92 \\ 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 158 factors with signed exponents of absolute value up to 11597937; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (4 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 169)y^3 + (-26s + 42585)y^2 + (-223s + 688385)y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 341y^8 + 84470y^7 + 5834003y^6 + 263122559y^5 + 7 \\ & 180070821y^4 + 140622190062y^3 + 1920411009606y^2 + 1444123966095 \\ & 9y + 76675889982213 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right)s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4632621291525 = 3 \cdot 5^2 \cdot 61768283887$:

$$\begin{pmatrix} 926524258305 & 726591467135 & 493991577802 & 319189311657 & 504916185110 & 743419938445 & 352533320822 & 749934338429 & 658490721791 & 737628397608 \\ 0 & 5 & 2 & 0 & 4 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 351 factors with signed exponents of absolute value up to 913638474; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 169)y^3 + (-26s + 42585)y^2 + (-223s + 688385)y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 341y^8 + 84470y^7 + 5834003y^6 + 263122559y^5 + 7180070821y^4 + 140622190062y^3 + 1920411009606y^2 + 14441239660959y + 76675889982213$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right)s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5610924323529 = 3 \cdot 7 \cdot 41 \cdot 271 \cdot 24047059$:

$$\begin{pmatrix} 5610924323529 & 2501272477059 & 3912479458208 & 3809475082778 & 4378846937718 & 1269781019844 & 360937245643 & 432259038917 & 2014765609147 & 4443095873087 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 394 factors with signed exponents of absolute value up to 2634332384; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s + 169)y^3 + (-26s + 42585)y^2 + (-223s + 688385)y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 341y^8 + 84470y^7 + 5834003y^6 + 263122559y^5 + 7180070821y^4 + 140622190062y^3 + 1920411009606y^2 + 14441239660959y + 76675889982213$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right) s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3224017064919 = 3 \cdot 7^2 \cdot 41 \cdot 9281 \cdot 57637$:

$$\begin{pmatrix} 460573866417 & 358499068724 & 139339085430 & 453081714576 & 406654364579 & 343315426363 & 179605807510 & 187420834880 & 350287547942 & 268376770279 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 7 & 1 & 0 & 6 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 359 factors with signed exponents of absolute value up to 9878942347; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 4)y^3 + (-8s - 37286)y^2 + (-163s + 361971)y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 7y^8 - 74580y^7 + 5323143y^6 - 74934384y^5 + 182980715y^4 \\ & - 3269841036y^3 + 219290760724y^2 - 2491338258864y + 8109734610 \\ & 432 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right) s, \quad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18644750208 = 2^7 \cdot 3^3 \cdot 7 \cdot 797 \cdot 967$:

$$\begin{pmatrix} 129477432 & 95902242 & 108570576 & 97337992 & 119631291 & 629817 & 97699846 & 38209097 & 2736289 & 22706110 \\ 0 & 6 & 0 & 4 & 5 & 4 & 0 & 0 & 4 & 4 \\ 0 & 0 & 6 & 4 & 0 & 4 & 2 & 5 & 4 & 4 \\ 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 138 factors with signed exponents of absolute value up to 27395003; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 4)y^3 + (-8s - 37286)y^2 + (-163s + 361971)y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 7y^8 - 74580y^7 + 5323143y^6 - 74934384y^5 + 182980715y^4 - 3269841036y^3 + 219290760724y^2 - 2491338258864y + 8109734610432$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right)s, \quad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4119743462400 = 2^{10} \cdot 3^6 \cdot 5^2 \cdot 31 \cdot 7121$:

$$\begin{pmatrix} 79470360 & 3739320 & 26734572 & 59723190 & 13670058 & 71995302 & 36562432 & 63543761 & 75496852 & 30112327 \\ 0 & 90 & 76 & 8 & 2 & 60 & 50 & 82 & 83 & 47 \\ 0 & 0 & 4 & 2 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 6 & 0 & 2 & 0 & 5 & 5 & 0 \\ 0 & 0 & 0 & 0 & 6 & 0 & 2 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 138 factors with signed exponents of absolute value up to 13045; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 4)y^3 + (-8s - 37286)y^2 + (-163s + 361971)y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 7y^8 - 74580y^7 + 5323143y^6 - 74934384y^5 + 182980715y^4 - 3269841036y^3 + 219290760724y^2 - 2491338258864y + 8109734610432$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right)s, \quad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2628553539584 = 2^{25} \cdot 7 \cdot 19^2 \cdot 31$:

$$\begin{pmatrix} 527744 & 48448 & 153216 & 252660 & 456196 & 353046 & 432639 & 292722 & 457558 & 294948 \\ 0 & 64 & 0 & 48 & 12 & 44 & 16 & 58 & 6 & 36 \\ 0 & 0 & 608 & 316 & 224 & 398 & 372 & 414 & 598 & 67 \\ 0 & 0 & 0 & 4 & 0 & 2 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 140 factors with signed exponents of absolute value up to 384994603; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (2 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.